

WEEK 1 – PROJECT PLANNING & SYSTEM ARCHITECTURE

College Skill Exchange Platform

Full Stack Web Application – Planning & Design Report

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Course	BCA
Task	Week 1 – Project Planning & System Architecture
Project Type	Full Stack Web Application
GitHub URL	To be added after project repository creation

1. Project Brief

The College Skill Exchange Platform is a full-stack web application designed to help college students discover, exchange, and develop skills with other students. A student can create a profile, list skills they can teach, add skills they want to learn, search for suitable students, and send collaboration or skill-exchange requests. The platform is intended to make peer-to-peer learning and project collaboration easier inside a college community.

The first stage of the project focuses on planning rather than implementation. This report defines the objectives, requirements, user flow, frontend wireframes, technical architecture, API structure, database design, and design rationale that will guide the later development stages.

2. Problem Statement

Students often have different technical, creative, communication, or academic skills, but there is no simple centralized platform where they can find classmates according to the skills they can teach or want to learn. This project aims to provide a structured digital environment for skill exchange, peer learning, and project collaboration.

3. Objectives

- Create a simple platform for students to exchange skills and knowledge.
- Allow users to maintain profiles containing skills they can teach and skills they want to learn.
- Provide skill-based student search and filtering.
- Enable students to send and manage collaboration requests.
- Support project collaboration between students with complementary skills.
- Design a scalable architecture that can later support messaging, notifications, and administration.
- Provide a clear foundation for frontend and backend implementation.

4. Target Users

- College students who want to learn new skills.

- Students who want to teach or share their existing skills.
- Students searching for teammates for academic or personal projects.
- Administrators who may manage users and reported content in future versions.

5. Key Features

Feature	Purpose
User Registration & Login	Create an account and securely access the platform.
Student Profile	Display name, bio, department, skills, learning interests, and project interests.
Skill Management	Add skills a user can teach and skills they want to learn.
Skill Search	Search students by skill, category, or learning requirement.
Exchange Request	Send, accept, reject, or cancel skill-exchange requests.
Project Collaboration	Create or join projects and find students with required skills.
Messaging	Planned feature for communication between matched students.
Notifications	Planned alerts for requests, messages, and project updates.

6. Functional Requirements

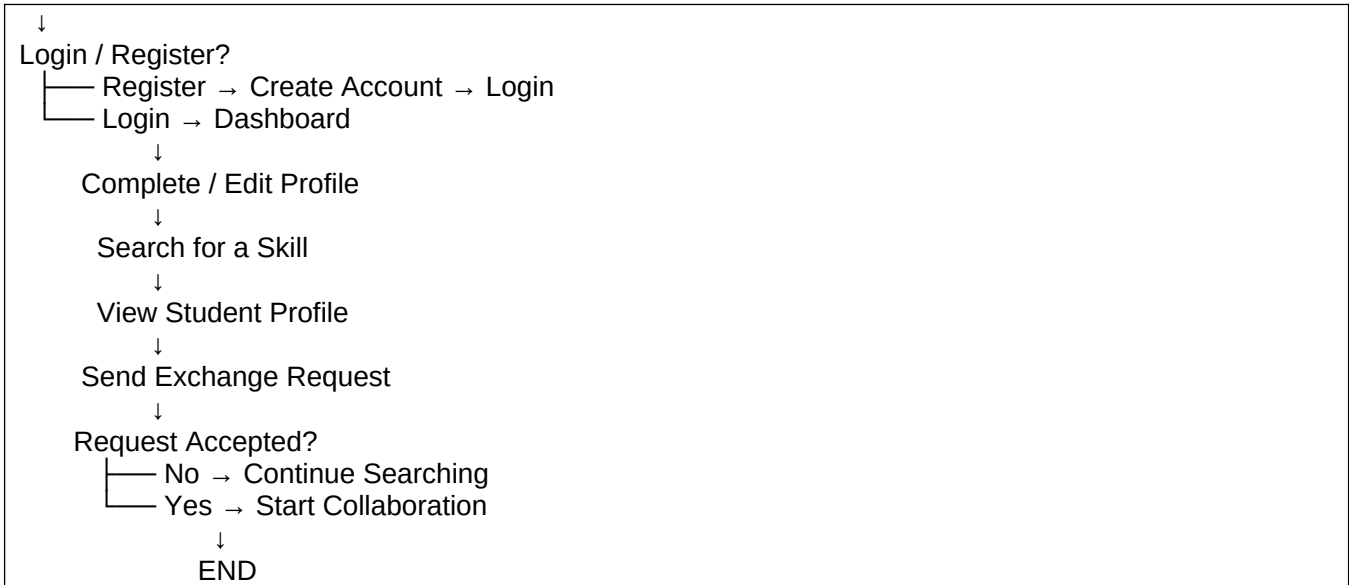
- The system shall allow a new student to register an account.
- The system shall allow registered users to log in and log out.
- Users shall be able to create and update their profiles.
- Users shall be able to add, edit, and remove skills.
- Users shall be able to search for students by skill.
- Users shall be able to send skill-exchange requests.
- Users shall be able to accept or reject incoming requests.
- Users shall be able to create and view collaboration projects.
- The backend shall provide REST API endpoints for frontend operations.
- The database shall store users, skills, projects, requests, and related records.

7. Non-Functional Requirements

- Usability: The interface should be simple and easy for college students to understand.
- Performance: Common pages and API requests should respond quickly.
- Security: Passwords should be securely hashed and authentication should protect private operations.
- Scalability: The architecture should allow additional features without major redesign.
- Maintainability: Frontend, backend, API, and database responsibilities should remain separated.
- Responsiveness: The frontend should work on desktop and mobile screen sizes.

8. User Interaction Flowchart





9. Frontend Wireframes

The following low-fidelity wireframes describe the planned screen structure. They can be converted into visual designs during the implementation phase.

A. Login Screen

COLLEGE SKILL EXCHANGE

Email:

Password:

[LOGIN]

New user? [CREATE ACCOUNT]

B. Dashboard

Logo | Home | Skills | Projects | Messages |

Welcome, Student!

[Search a skill...] [Search]

My Skills

[Python]

[C/C++]

Skills I Want to Learn

[React] [UI/UX]

Recommended Students

[Profile] [Skill] [View Profile]

C. Student Profile

← Back

Student Profile

[Profile Photo]

Name: Rahul
Department: BCA

Can Teach: Python, C, HTML
Wants to Learn: JavaScript, React

[SEND EXCHANGE REQUEST]

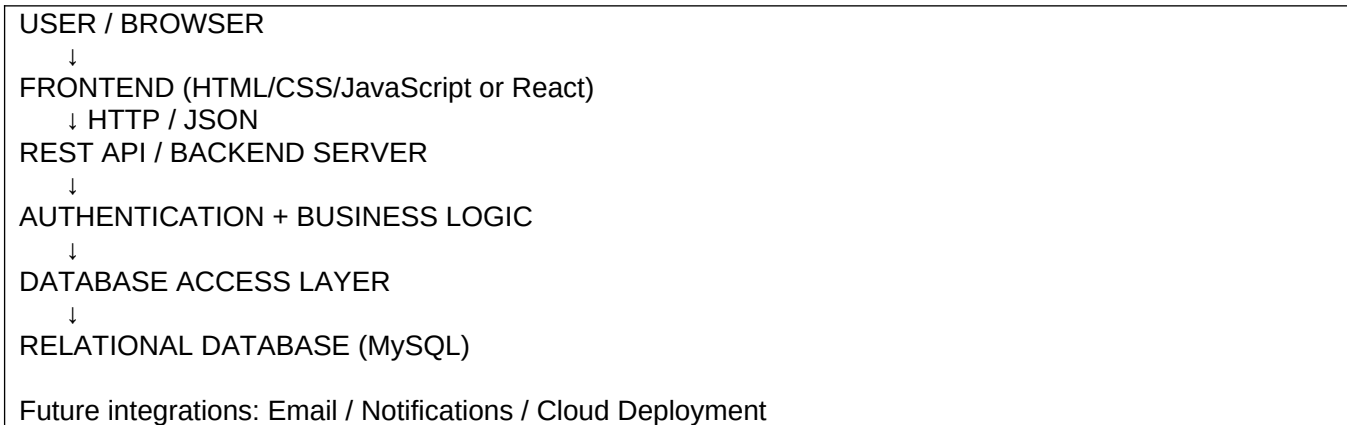
D. Project Collaboration

Projects

[Create Project]

Project: Student Expense Tracker
Required: Python | UI/UX
Members: 3
[VIEW / JOIN]

10. Technical Architecture



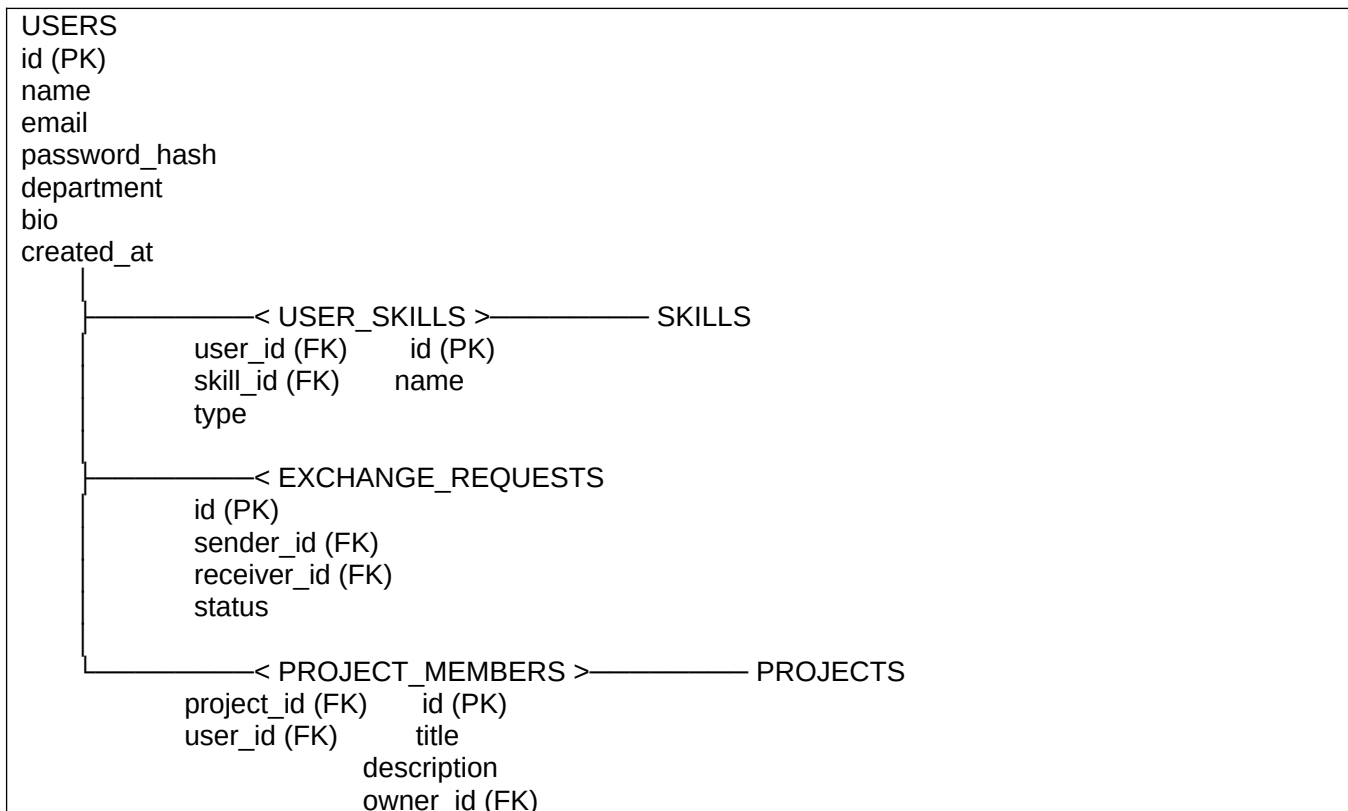
11. Technology Stack

Layer	Planned Technology
Frontend	HTML, CSS, JavaScript; React can be used in the implementation phase.
Backend	Node.js + Express.js (planned).
API	REST API using JSON.
Database	MySQL relational database.
Authentication	JWT-based authentication with securely hashed passwords.
Development Tool	Visual Studio Code.
Version Control	Git and GitHub.

12. Backend Operations & API Endpoints

Endpoint	Operation
POST /api/auth/register	Register a new user.
POST /api/auth/login	Authenticate user and return a session/token.
GET /api/users/:id	Get a student profile.
PUT /api/users/:id	Update profile information.
GET /api/skills	Get available skills.
POST /api/skills	Add a skill to a user's profile.
GET /api/users?skill=python	Search students by skill.
POST /api/requests	Send a skill-exchange request.
GET /api/requests	View incoming/outgoing requests.
PUT /api/requests/:id	Accept or reject a request.
POST /api/projects	Create a collaboration project.
GET /api/projects	List available projects.

13. Database Schema / ER Design



14. Database Tables

Table	Important Fields
users	id, name, email, password_hash, department, bio, created_at
skills	id, name, type
user_skills	user_id, skill_id
exchange_requests	id, sender_id, receiver_id, status, created_at
projects	id, owner_id, title, description, created_at
project_members	project_id, user_id, role
messages (future)	id, sender_id, receiver_id, message, created_at

15. Design Decisions & Rationale

- A simple dashboard is selected so students can quickly search skills and see relevant users.
- A relational database is planned because users, skills, requests, and projects have clear relationships.
- REST APIs separate frontend presentation from backend business logic and make future mobile-app integration easier.
- JWT authentication is planned to protect user-specific operations.
- A responsive frontend is preferred because students may access the platform from both computers and phones.
- The architecture is modular so features such as notifications, messaging, ratings, and administration can be added later.

16. Security Considerations

- Passwords will not be stored as plain text; secure password hashing will be used.
- Authentication tokens will be required for protected API operations.
- Backend validation will be applied to user input.
- Database queries will use safe parameterized methods to reduce injection risks.
- Users will only be allowed to modify resources for which they have permission.

17. Future Scope

- Skill ratings and student reviews.
- Real-time chat and notifications.
- AI-based student/skill matching.
- Project recommendation system.
- College-specific communities.
- Admin dashboard and reporting system.
- Cloud deployment and mobile application.

18. GitHub Plan

The project source code will be maintained in a GitHub repository. The repository will contain frontend, backend, database documentation, README instructions, and future implementation work.

GitHub Project URL: https://github.com/Tanishkchauhan75/-college-skill-exchange-platform.git

19. Conclusion

The Week 1 planning stage establishes the foundation for the College Skill Exchange Platform. The project requirements, user flow, wireframes, architecture, API structure, and database schema have been defined before implementation. This approach reduces development confusion and provides a clear roadmap for building the frontend and backend in subsequent weeks. The proposed architecture is simple enough for a BCA-level project while remaining scalable for additional features.

20. Week 1 Deliverables Checklist

- Completed – Project concept and brief
- Completed – Objectives and requirements
- Completed – User interaction flowchart

- Completed – Frontend wireframes
- Completed – Technical architecture
- Completed – Backend/API plan
- Completed – Database schema and relationships
- Completed – Design rationale
- Completed – GitHub repository plan